

SECTION 11 - PLANNED MAINTENANCE ASSIST

NOTE: Section 11 is specifically for systems at or above release 11.X.
For pre-11.X systems, refer to Section 12 of the PM manual.

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General Requirements

Personnel Requirements:	1 Field Engineer for 2 hours
Tools and Test Equipment:	LVShim Phantom Assembly Nesting Plate Assembly DQA Phantom Head Coil Body Sphere and Loader Service Key
Replacement Parts:	None
Consumables:	None
Safety:	Standard Magnet Safety Procedures for personnel working in and around magnets.
Required Conditions:	System release 11X or higher Completion of a successful LVShim is a prerequisite to performing the SPT procedure. (See the first section of this procedure.) For more information about LVShim, see <i>Direction 2333500, Signa Excite Service Methods</i> , Adjust and Cal / System level Procedures / LVShim.

Objective

The goal of PM Assist is to ensure that the Planned Maintenance (PM) is done in a timely manner, is completed, and problems found are resolved quickly. Install In Spec was developed to ensure that the scanner performance after installation and initial calibration meets GEMS quality standards before turning the MR scanner over to the customer. PM Assist was developed to ensure the quality of ongoing scanner maintenance during the lifetime of the scanner.

Flow of PM Assist

A typical PM session using the PM Assist feature will include the field engineer (FE) running SPT in PM mode and LVShim in Gradient Shim Mode, resolving any issues found, and then completing other tasks scheduled for the PM. Before leaving, the FE will run the PM Verification tool to ensure that the required tasks (SPT and LVShim) are recognized by the system as having been performed. The FE will also be warned of any issues outstanding and reminded that they need to be resolved promptly.

Theory of Operation

PM Assist is designed to perform three checks:

- **Check if a PM is due or overdue:** A PM should be done every other calendar month.
- **Check a PM for completeness.** This is defined as LVShim and SPT (in PM mode) completed within 24 hours of each other (it is recommended to run LVShim first). It does not matter whether the tests pass or fail. As long as both tools have been run, the PM is considered complete.
- **Check for severe errors remaining from the PM.** A severe error is an error that is expected to impact clinical image quality. LVShim and SPT PM mode results are checked for this type of errors. You have 21 days from the date the PM was completed to fix the severe errors.

After loading the Excite-II software, you must run the **PM Check** tool, and schedule your first PM in order to start PM Assist.

More Details

PM Due / Overdue

- If you did a PM sometime in January, you are expected to do the next PM sometime in March. The PM is considered due starting March 1 and overdue starting April 1.
- While the PM is due, a pop-up displays at each reboot of the scanner. You can avoid this pop-up by scheduling the PM for a specific date (within the due month). If the PM is not completed at that time, the pop-up will start showing up.
- Another pop-up displays at reboot when the PM is overdue. You must complete the PM in the month due to avoid this pop-up.

PM Completeness

- The latest SPT PM Mode results file marks the time of the PM. LVShim must be run within ± 24 hours of that time for the PM to be considered complete. If you do not have an SPT PM mode results file and an LVShim file within 24 hours of each other, the PM is incomplete and you must run both tools again, within 24 hours of each other.
- Each run of SPT PM Mode is considered the marker for a PM and will restart all the checks above. Only run SPT PM Mode at PM time, not between PMs.

Severe Errors Remaining from PM

- Any severe errors in the latest SPT PM Mode results file and the results from the last LVShim run within 24 hours of the SPT PM Mode run will be identified. PM Assist will then search through all subsequent SPT and LVShim files to see if the test(s) that failed passed at a later time. Once PM Assist finds a pass status for a test, it will mark it as successful. It does not matter if the test(s) fail later. Once a test has passed at or after a PM, it is considered successful, even if it fails later.
- You should NOT run SPT PM Mode again after fixing severe errors. Use SPT Full Mode with one or more of the failing test(s) selected.
- It is not necessary to have all the failing tests pass in the same SPT results file. PM Assist will search through all SPT files acquired after the PM for passing results. This is different from current Install In Spec behavior.
- If you have not fixed all severe errors within 21 days after the PM was completed, a pop-up will display at each reboot of the scanner.

Other Useful Information

- Run **PM Check** to mark the PM as completed.
- Run **PM History** to see the current status of PMs.
- View pm_assist.log, using "gedit" editor.

PM Assist Procedure

1. With the LVShim Phantom properly placed and land marked, the system must be configured to make the appropriate LVShim scans. Click on the Scan icon to launch the Patient Information window shown in Figure 1.

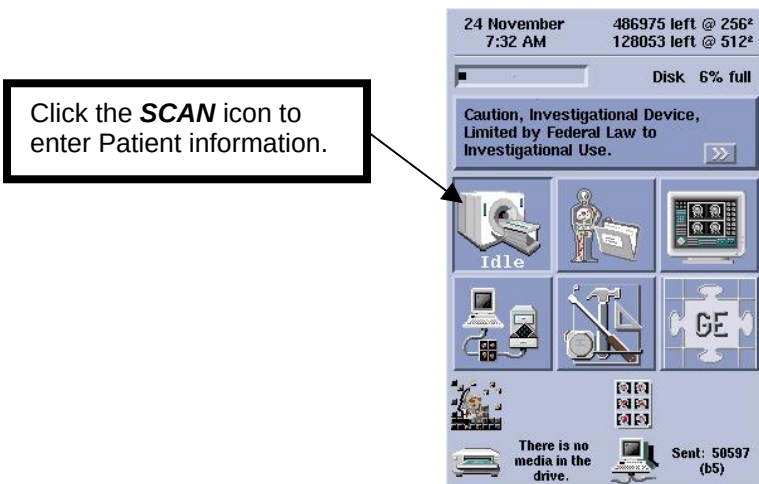


Figure 1

2. The Patient Information window will launch with the Patient Protocols section shown in Figure 2.

Figure 2

- To run the LVShim procedure, it is necessary to insert "geservice" in the Patient ID field and "150" in the Weight Lb field. Pick a Patient Name that will be easy to find in the browser for the Shim Test. (*Shim PM Test* is used for this example.) See Figure 3.

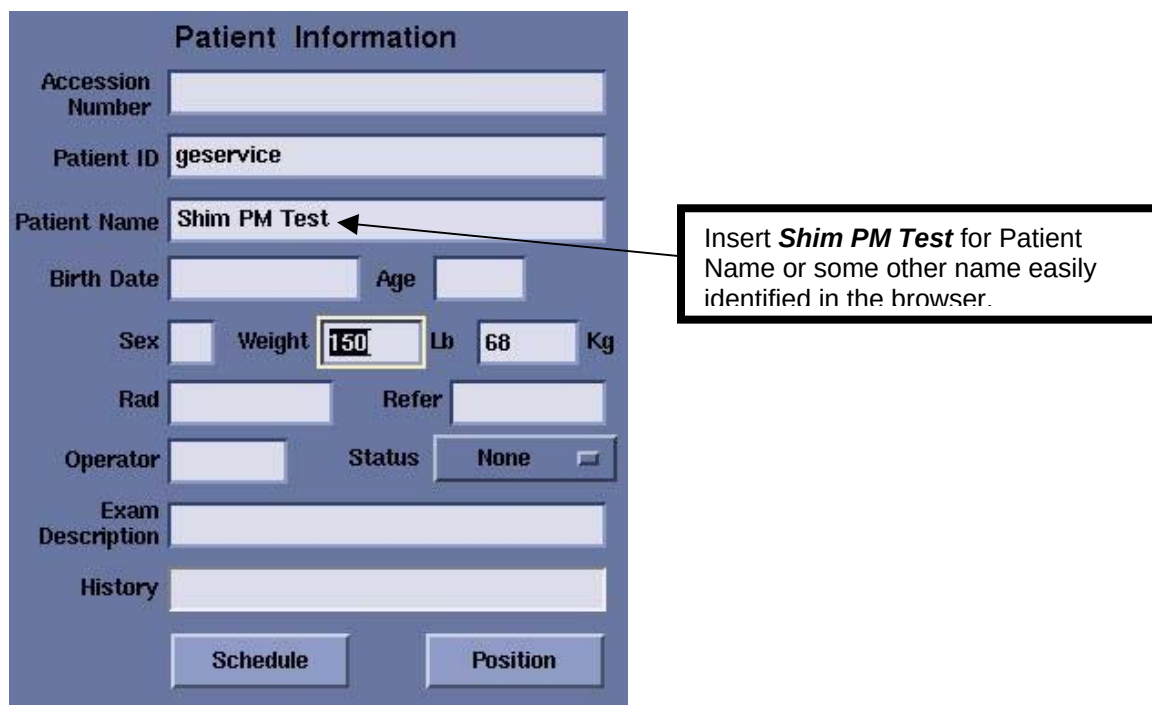


Figure 3

- From the pull-down box in the upper right of the Patient Protocols window, select "Service" shown in Figure 4.

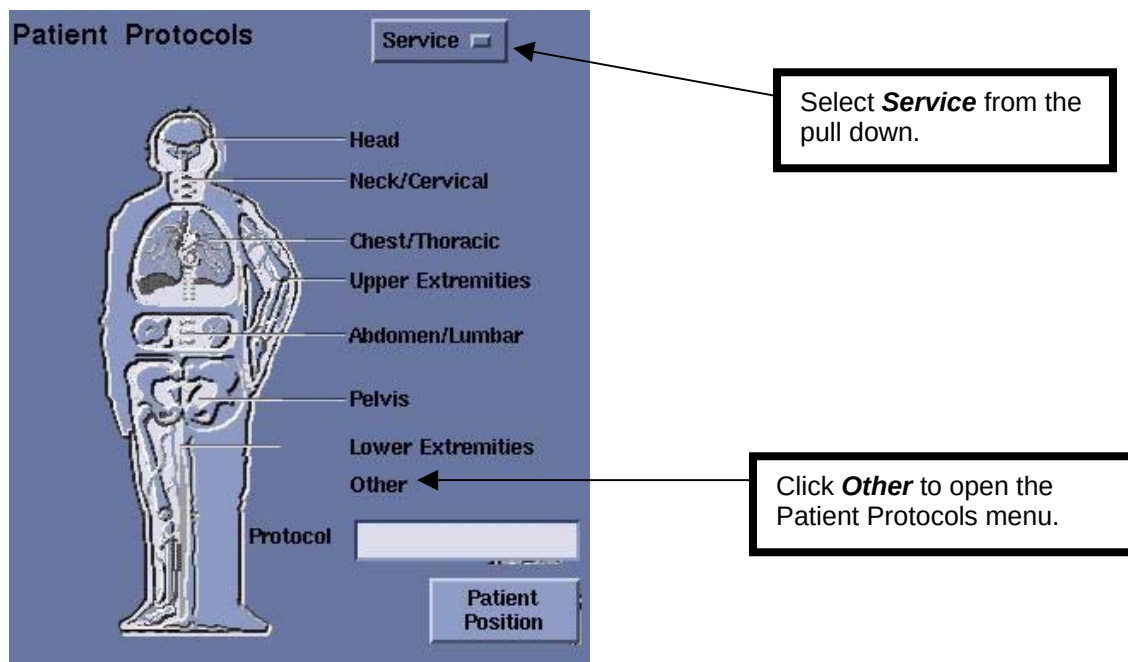


Figure 4

5. From the Protocol selection box, select "0.19 LVshim" in the Protocol column and "LVshim Localizer, Cal:Syst" in the Series column shown in Figure 5.
6. Click "Accept" (shown in Figure 5) when the selections are correct and complete.

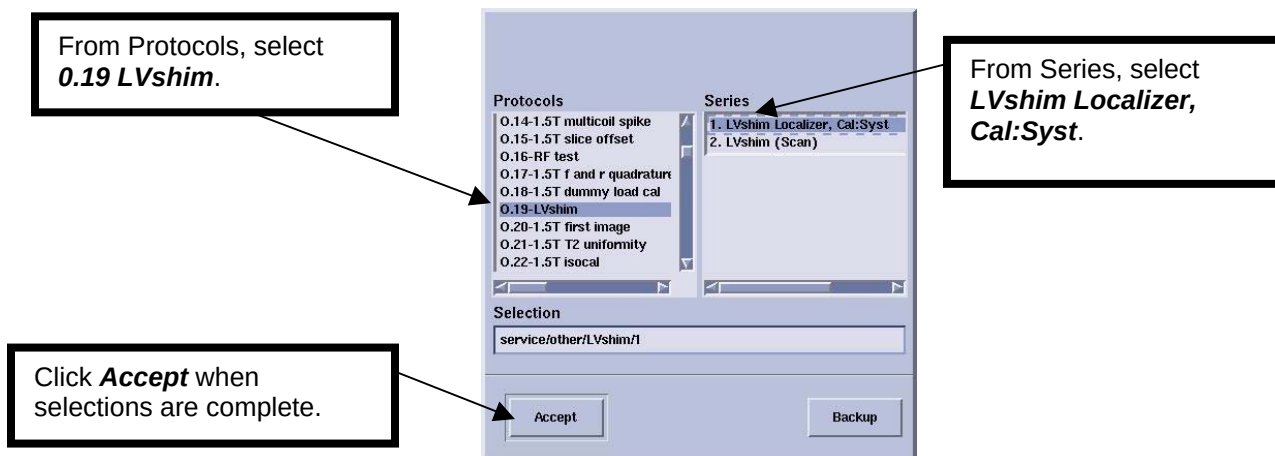


Figure 5

7. When the pop-up message: *A change has been made to Patient Position, Patient Entry, and/or coil. Please verify this selection and continue*, displays in the Patient Information section of the console, click "Apply All" to continue. See Figure 6.

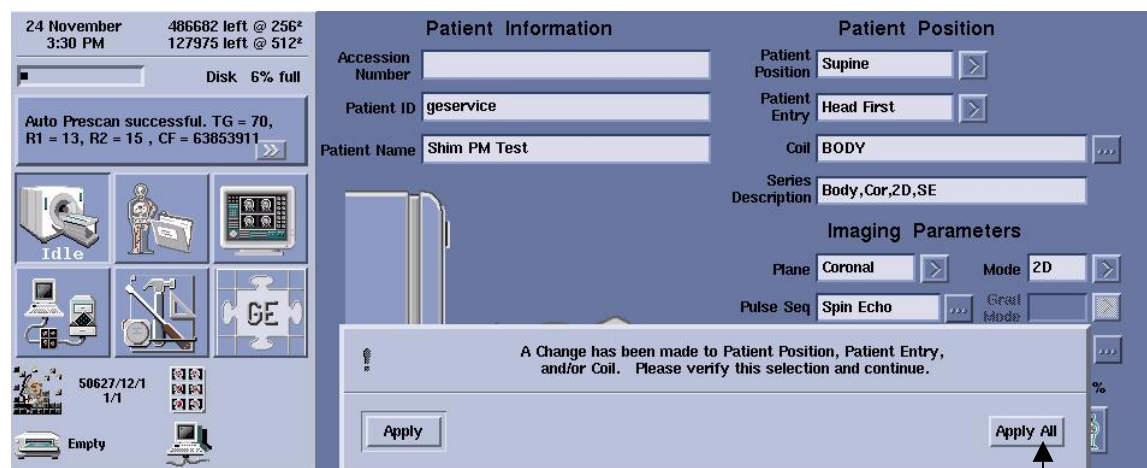


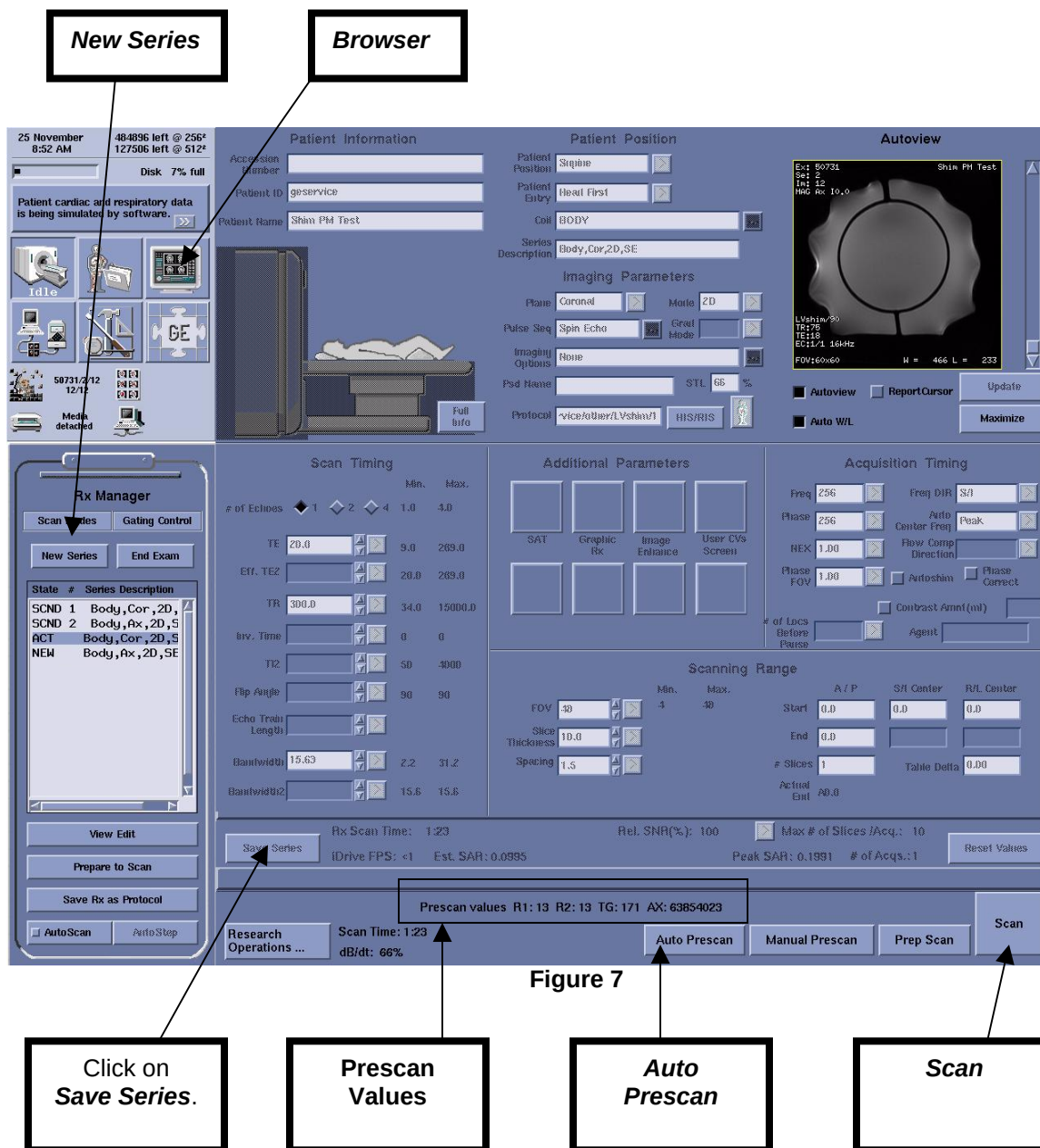
Figure 6

Note: For TwinSpeed, when a pop-up displays: *Whole or Zoom ??*, select "Whole."

8. From the Scan Prescription screen, click “Save Series.” See Figure 7.
9. From the Scan Prescription screen, click “Auto Pre-scan,” and verify the Prescan values are within an acceptable range: **R1** = 13 ±2, **R2** = 13 ±2, **TG** 151-171. See Figure 7.

Note: If prescan is not within the acceptable range, resolve this issue before continuing this procedure.

10. Click “Scan” on the Scan Prescription screen (shown in Figure 7) to initiate the LVshim Localizer Cal:Sys scan.



11. To access the browser, click the Browser icon shown in Figure 7. The browser window (shown in Figure 8) will launch.
12. Locate the image created when the "LVshim Localizer, Cal:Syst" Scan was performed in the browser. Find the Patient name in the Examinations column (left column) by looking for the name entered in Step 5. (This example used *Shim Pm Test*.) Click on "Shim Pm Test." The scans will now display in the right column. See Figure 8.
13. Click on the appropriate series in the right column. If a new name was used in Step 5, there will be only one series displayed in the right column (one image with the description Body,Cor,2) at this time. See Figure 8.

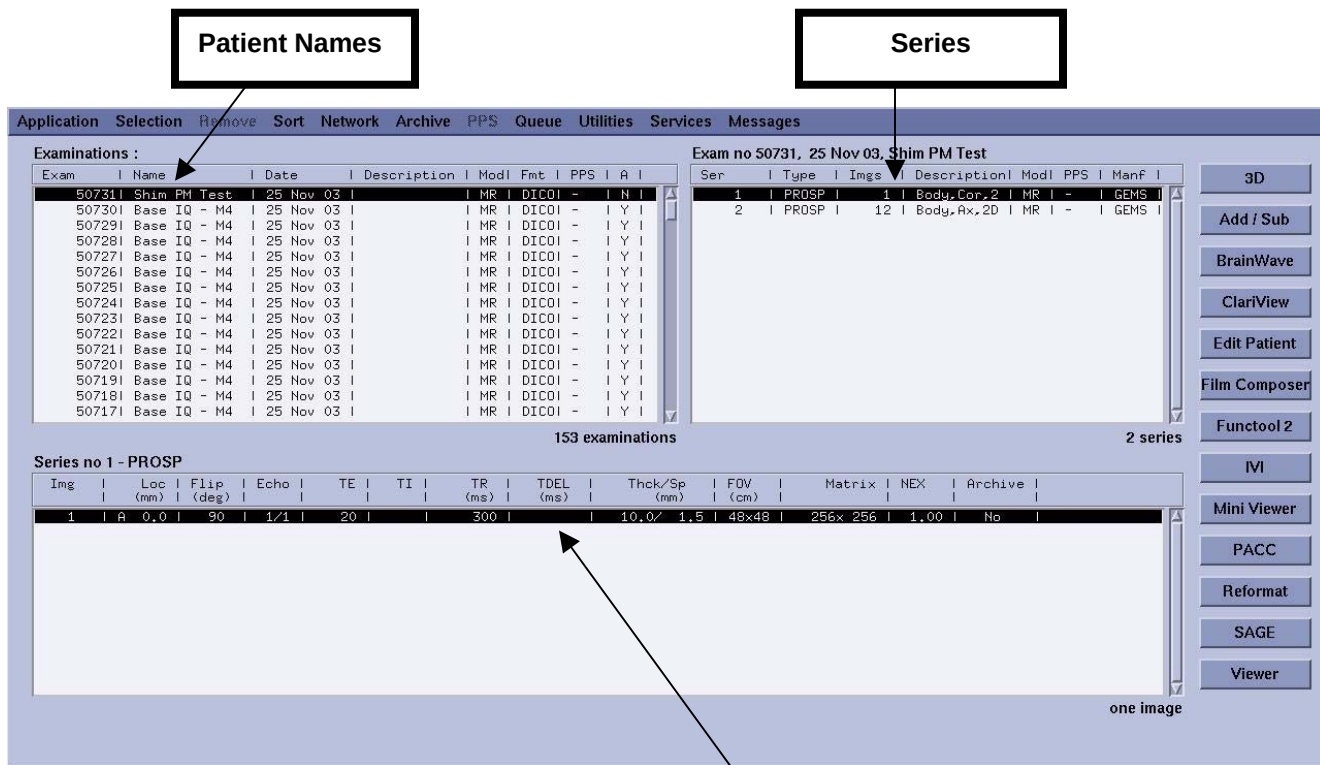


Figure 8

14. Double click on the image line in the lower window (shown in Figure 8) to automatically open the image shown in Figure 9.

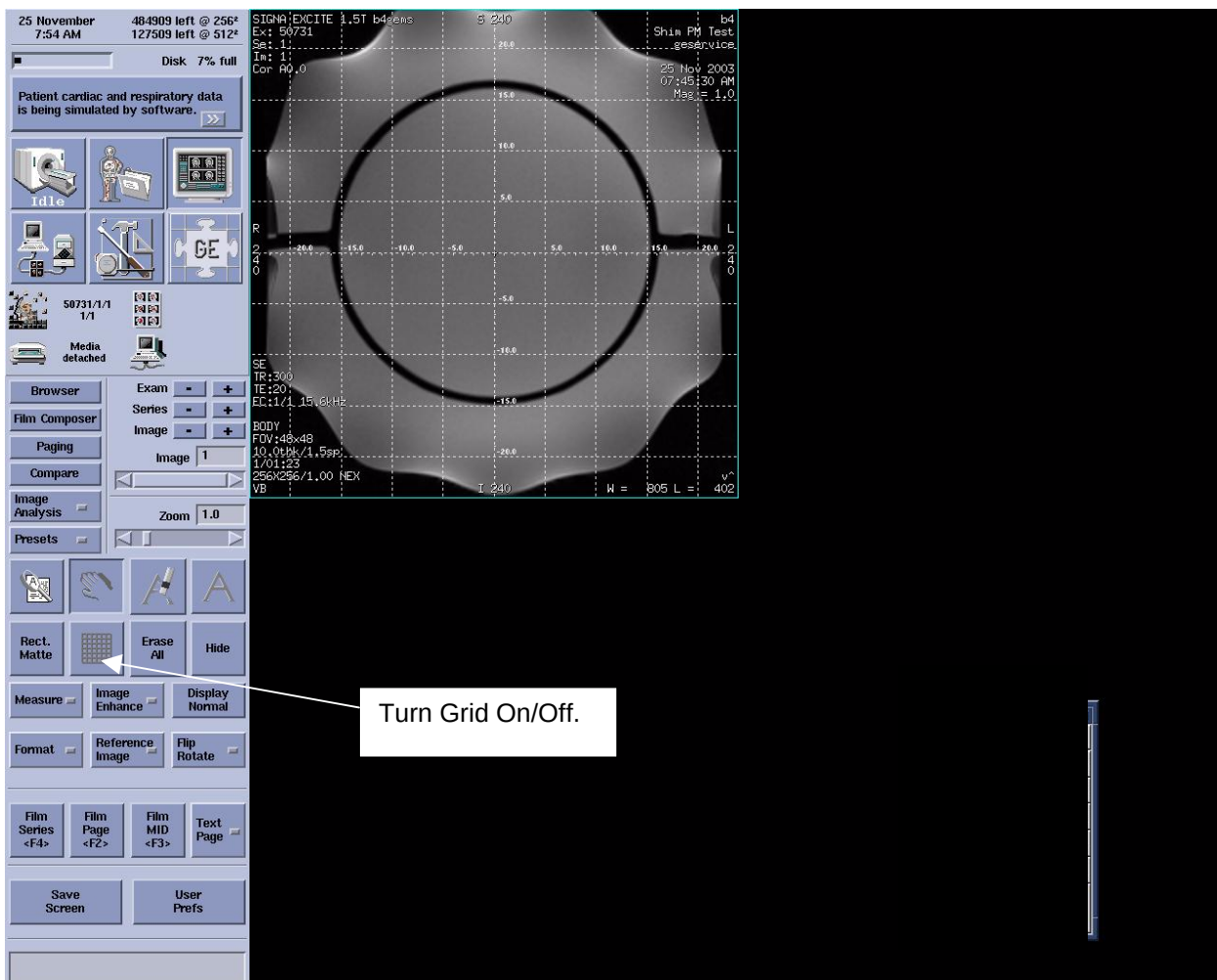


Figure 9

15. It is important to center the LVShim phantom in the Magnet bore. As seen in this example, make sure the Phantom is moved slightly toward the front of the Magnet (toward the Table). See Figure 9. The goal is to overlap the center of the phantom and the display grid.
16. Reposition the phantom and run the "LVshim Localizer, Cal:Syst" until the Phantom is properly centered.
17. Now that the LVShim Phantom is properly centered, the LVShim scans can be acquired. Return to the Patient Protocol window by clicking the Scan icon shown in Figure 1.
18. Click "New Series" on the Rx Manager shown in Figure 7.

19. Click “Other” in the Patient Protocols window shown in Figure 4. A Protocol selection box (shown in Figure 10) will open.

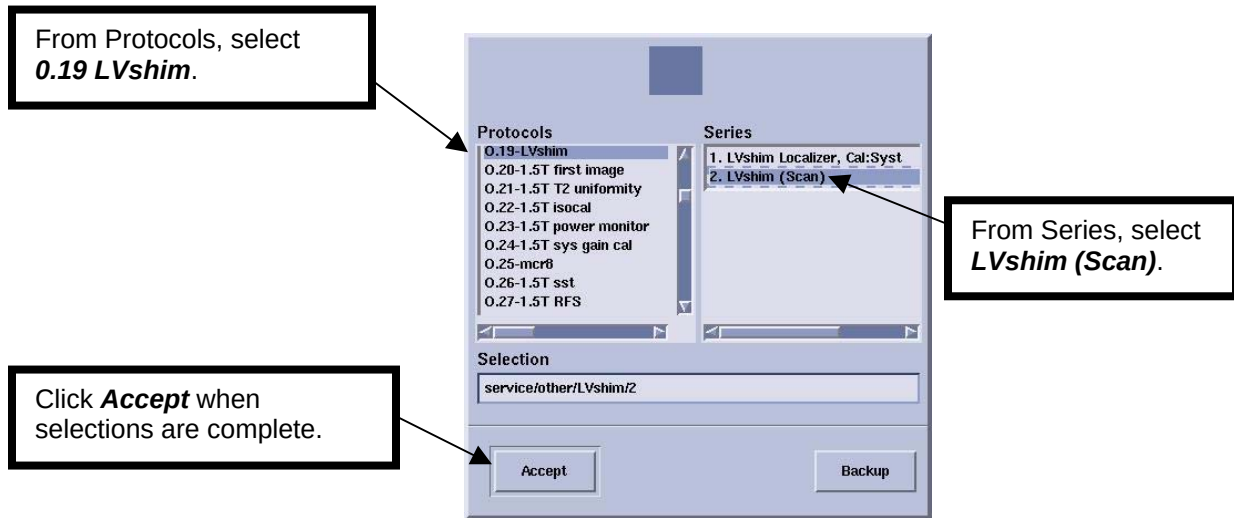


Figure 10

20. From the Protocol selection box, select “0.19 LVshim” in the Protocols column and “LVshim (Scan)” in the Series column. See Figure 10.
21. Click “Accept” (shown in Figure 10) when the selections are correct and complete.

22. Start the LVShim Scan, and click in the following order: Save Series => Prepare to Scan => Scan.
See Figure 11.

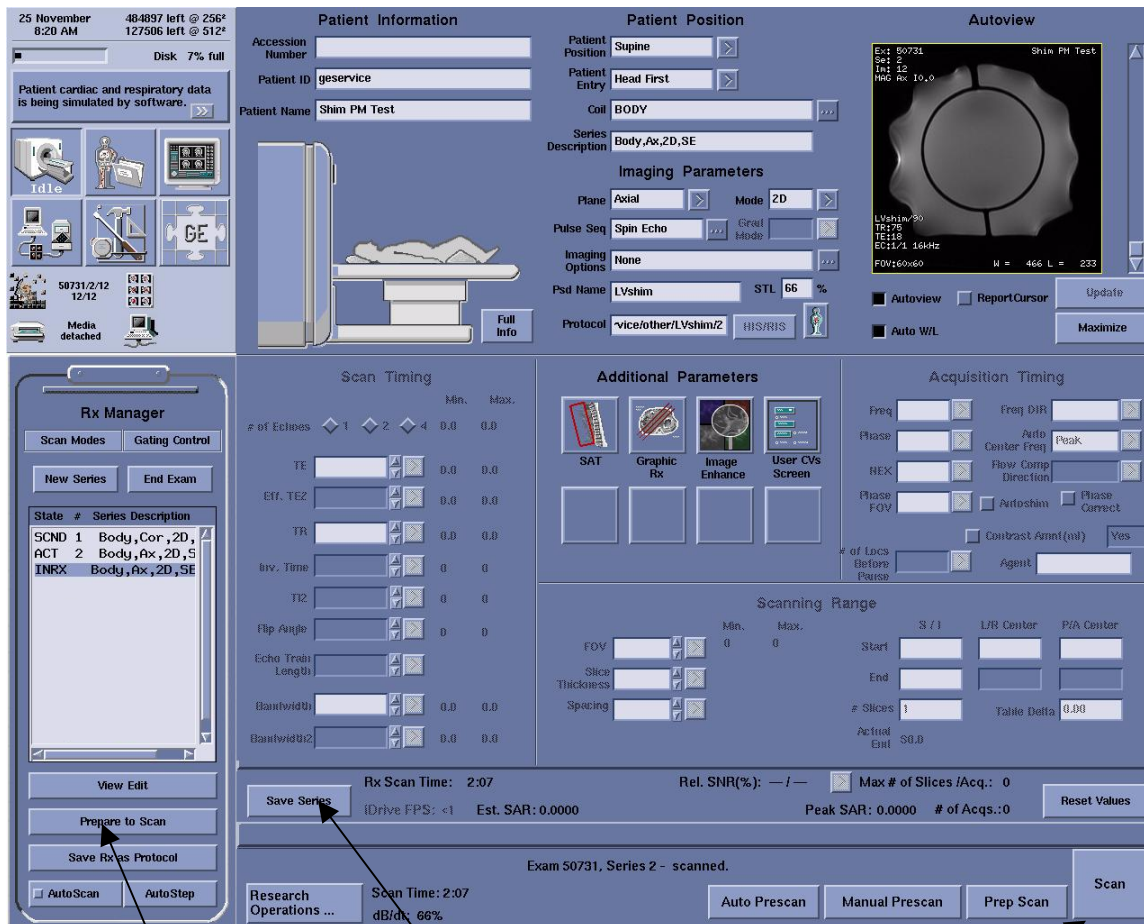


Figure 11

2: Click

1: Click

3: Click

23. At the completion of the scan sequence, proceed to the LVShim tool to verify results (next step).

24. Access the Common Service Desktop by clicking on the Tools icon shown in Figure 12.

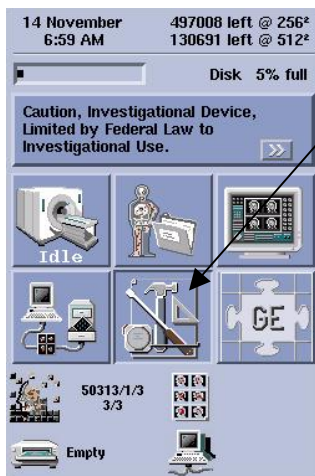


Figure 12

25. When the Common Service Desktop home page shown in Figure 13 opens, click on the PM icon.

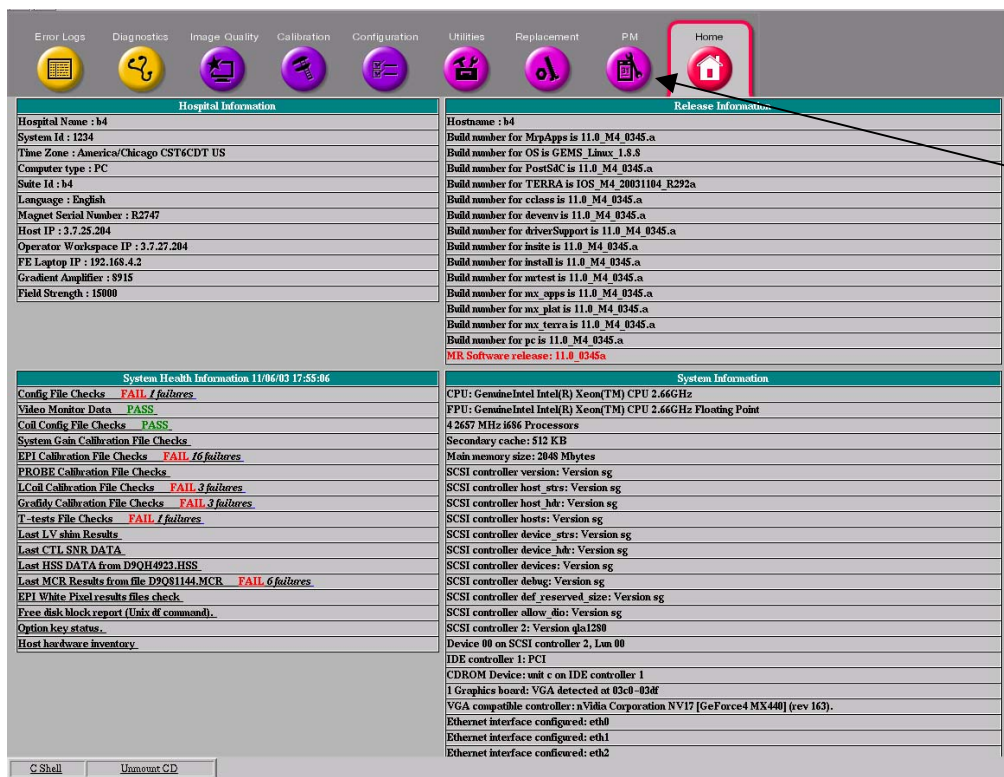


Figure 13

26. Click on the PM folder in the left column of the PM main page to expand the folder and show the PM selections. See Figures 14a, 14b, 15a and 15b.

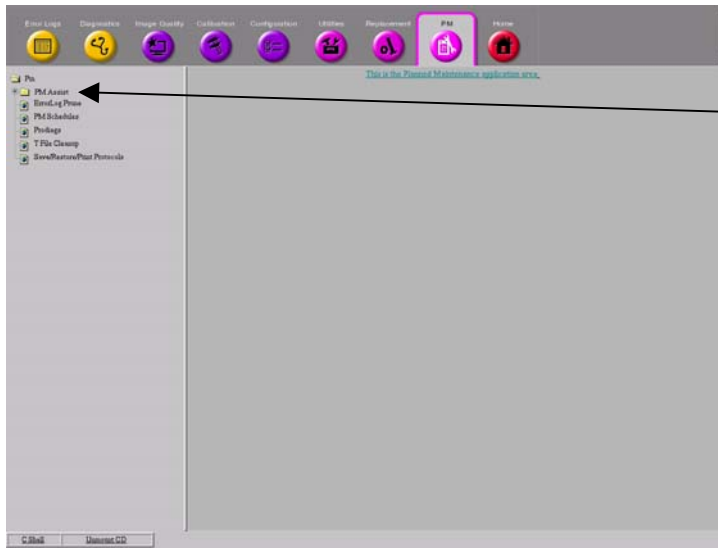


Figure 14a

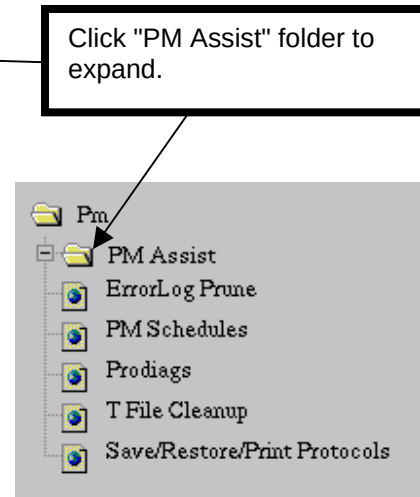


Figure 14b

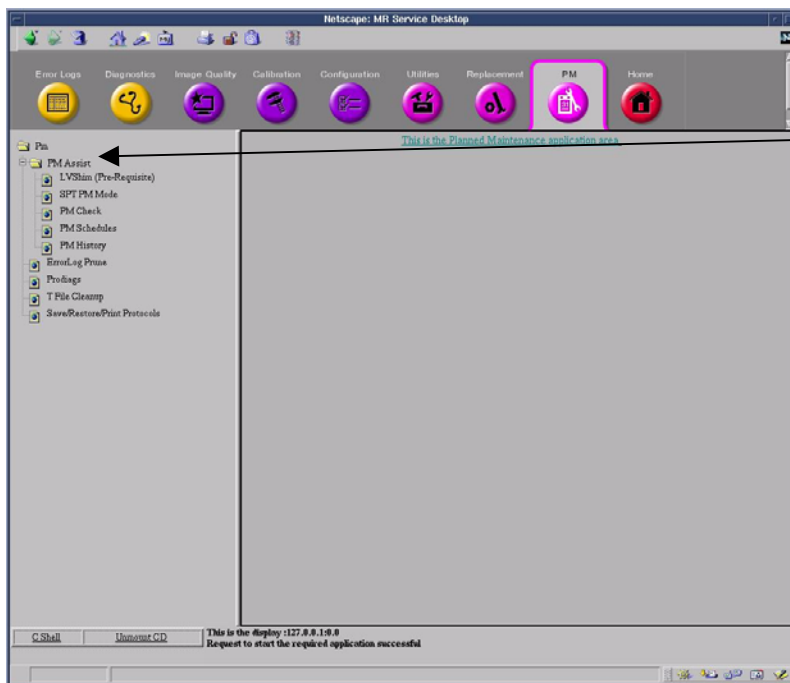


Figure 15a

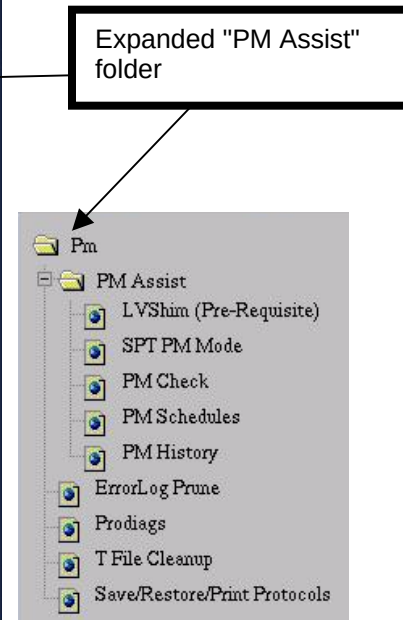


Figure 15b

LVShim Tool

1. Click on "LVShim (Pre-Requisite)" in the expanded PM Assist folder shown in Figure 16b. The right side of the screen will provide more details about LVShim and a link to start the LVShim PM tool.

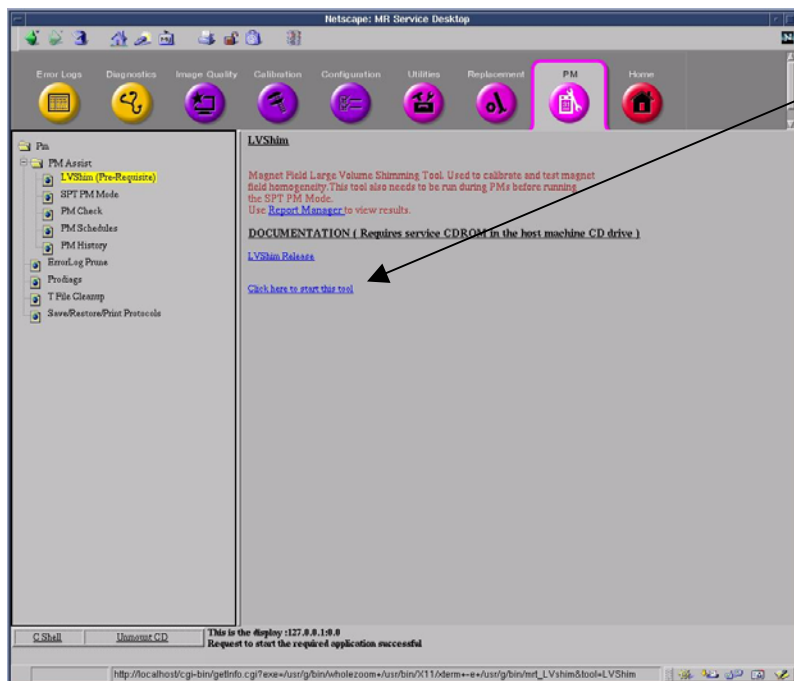


Figure 16a

Click on this link to start LVShim tool.

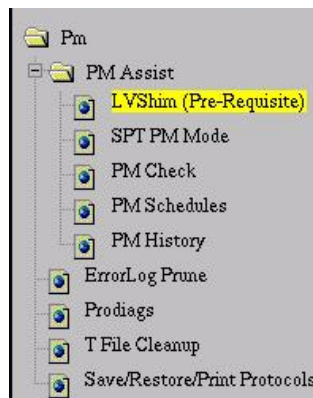


Figure 16b

2. Click on the link: "[Click here to start this tool](#)" (shown in Figure 16a) to start the LV Shim tool and open the LVShim page shown in Figure 17.

Note: For TwinSpeed, when a pop-up displays: *Whole or Zoom ??*, select "Whole."

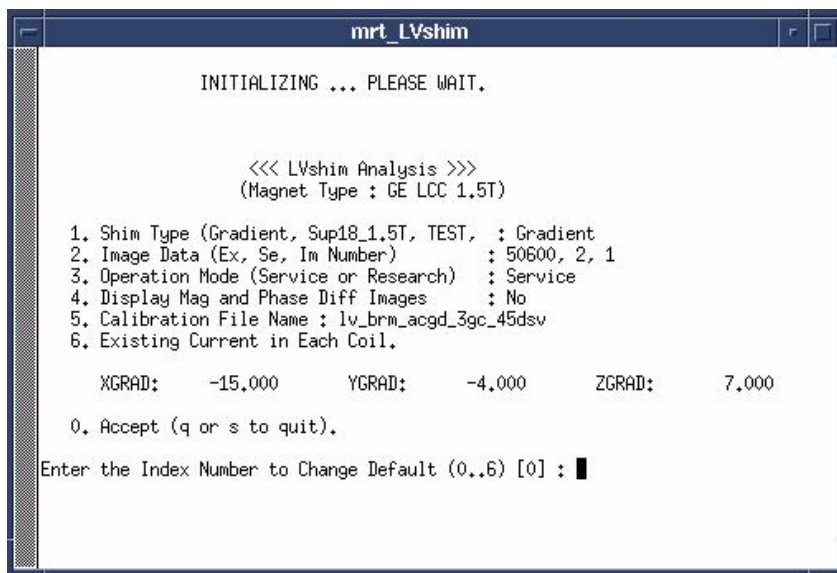


Figure 17

Default values for LVShim analysis tool:

- **Shim Type** = Gradient
- **Image Data** = This will auto fill with the latest LVShim images.
- **Operation Mode** = Service
- **Display Mag and Phase Diff Images** = No
- **Calibration File** = lv_brn_acgd_3gc_45dsv
- **Existing Current In Each Coil**

Note: Verify the image data matches the exam series image scanned in Step 25.

3. To start the analysis, press the [Enter] key. The results will be returned as follows:

(1, 0): -137.67 [10]. The Z1 limit is 10; the actual measure value is -137.67.

As shown in Figure 18, Grad Shim has not passed.

Note: Proceed to Step 9 if the message displayed is: *Grad Shim Passed*.

LVShim Harmonic Correlation:

1,0 = Z1, 2,0 = Z2, 3,0 = Z3, 4,0 = Z4, 5,0 = Z5
 6,0 = Z6, 1,1 = Y, 1,-1 = X, 2,1 = ZY, 2,-1 = ZX
 2,2 = X2-Y2, 2,-2 = XY, 3,1 = Z2Y, 3,-1 = Z2X
 3,2 = Z(X2-Y2), 3,-2 = ZYX, 3,3 = Y3, 3,-3 = X3

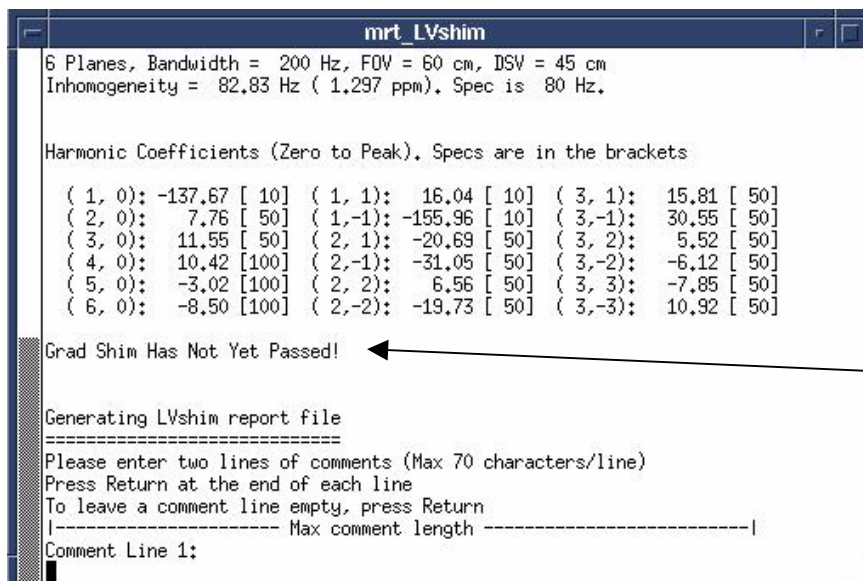


Figure 18

Status:
"Grad Shim Not Yet Passed"

4. Entering comments is allowed, but not required. Press the [Enter] key to continue.

5. To accept the new Grad Shim values, type “y” and press the [Enter] shown in Figure 19.

The screenshot shows the 'mrt_LVshim' window. At the top, it displays a grid of coordinates and values: (5, 0): -3.02 [100], (2, 2): 6.56 [50], (3, 3): -7.85 [50], (6, 0): -8.50 [100], (2,-2): -19.73 [50], (3,-3): 10.92 [50]. Below this, it says 'Grad Shim Has Not Yet Passed!'. Then, it prompts for a report file: 'Generating LVshim report file', followed by instructions to enter two lines of comments (Max 70 characters/line). Below the comment lines, it shows 'Shim Current Settings' with a table:

Coil	Existing	Delta	New
XGRAD	-15,000	14,000	-1,000
YGRAD	-4,000	-1,000	-5,000
ZGRAD	7,000	-13,000	-6,000

At the bottom, it asks 'Accept the New Currents? (Y,N) [Y] : y'.

Figure 19

6. When the tool asks *Exit LVShim?*, type “n” and press the [Enter] key. See Figure 20.

This screenshot is similar to Figure 19, showing the same coordinate grid and shim current settings table. However, at the bottom, the prompt is 'Exit LVshim? (Y,N) [N] : n', indicating the user has chosen to exit the program.

Figure 20

7. Return to the Scan Prescription page, and click the [Scan] button. When scans are complete, return to the LVShim tool and press the [Enter] key. See Figure 21.

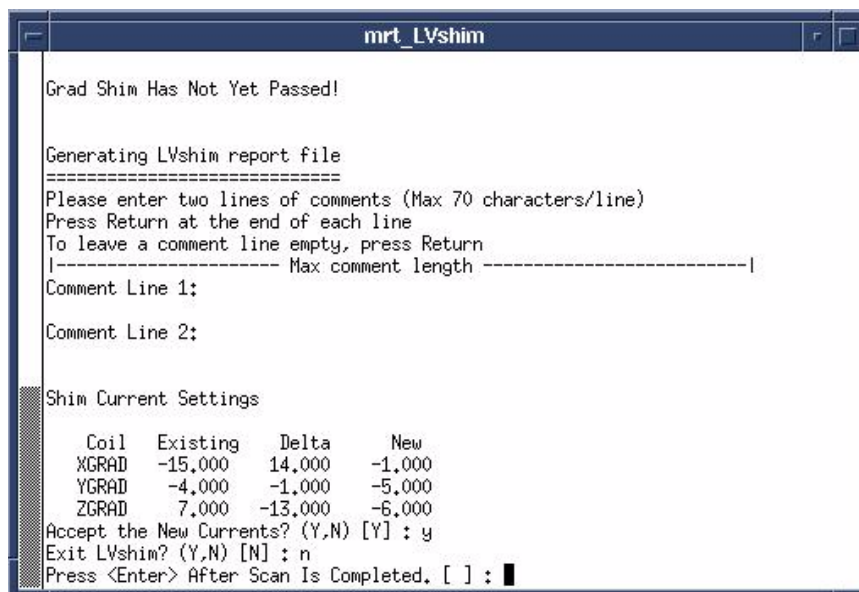


Figure 21

8. The LVShim tool will automatically select the 12 new images created by the last scan. Press the [Enter] key to start the analysis. See Figure 22.

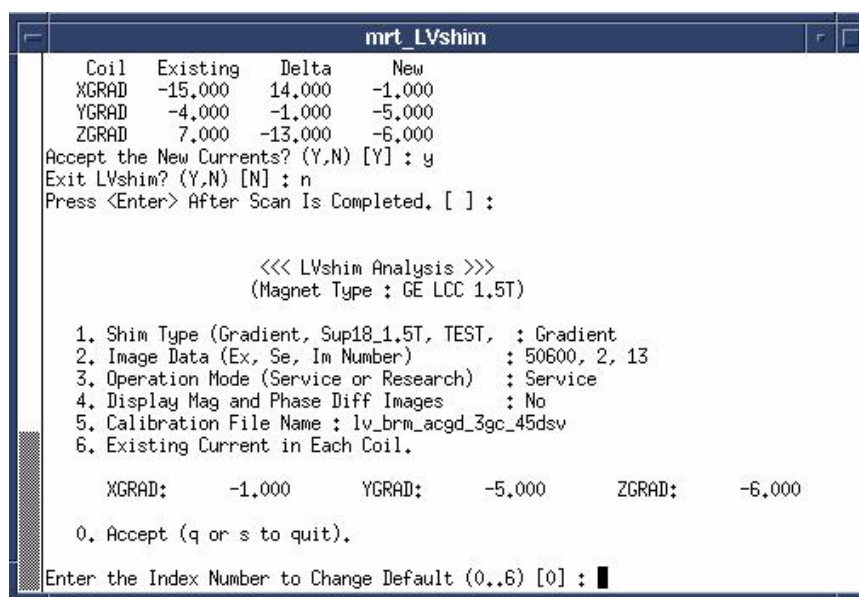
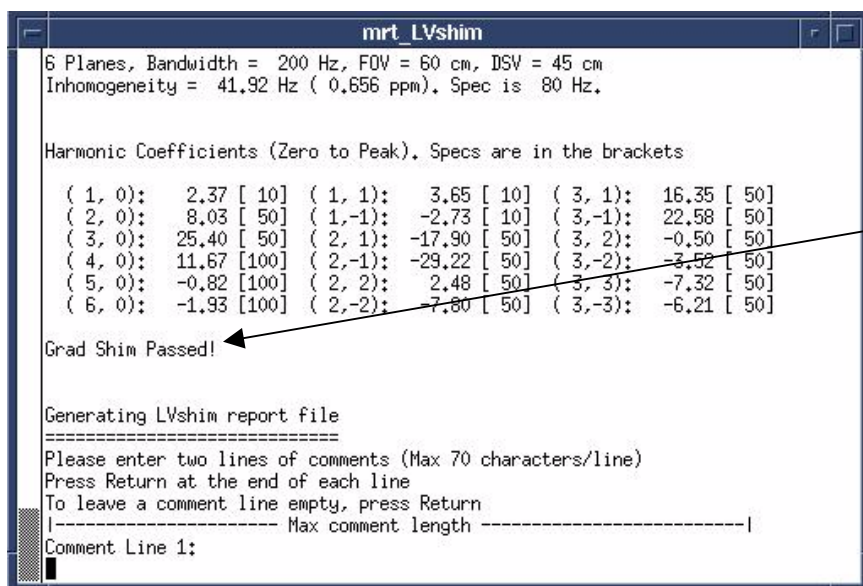


Figure 22

Note: Grad Shim is capable of reducing the X (1,-1) Y (1,1) and Z (1,0) harmonics. If any of the higher order harmonics are out of specification, a Supercon Shim may be required.

9. In the example shown in Figure 23, LVShim has now passed. Depending on the System, it may take only one pass or it may take several. Repeat the procedure until the system passes the LVShim.
10. Enter any applicable comments, and press the [Enter] key. See Figure 23.



Status:
"Grad Shim Passed"

Figure 23

11. To reject the new current values, enter "n" and press the [Enter] key. See Figure 24.

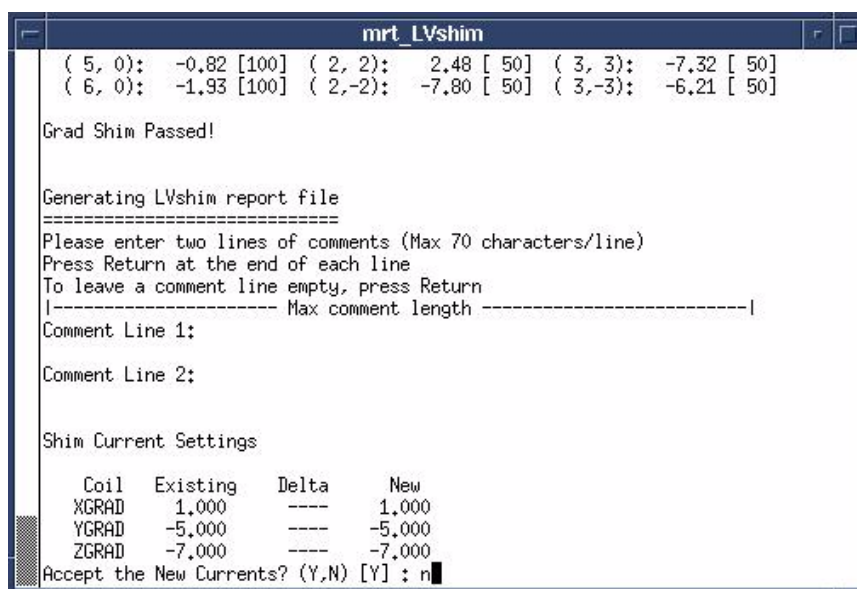


Figure 24

12. When the tool asks *Exit LVShim?*, type “y” and press the [Enter] key. See Figure 25.

```

mrt_LVshim
( 6, 0): -1.93 [100] ( 2,-2): -7.80 [ 50] ( 3,-3): -6.21 [ 50]
Grad Shim Passed!

Generating LVshim report file
=====
Please enter two lines of comments (Max 70 characters/line)
Press Return at the end of each line
To leave a comment line empty, press Return
|----- Max comment length -----|
Comment Line 1:
Comment Line 2:

Shim Current Settings
      Coil   Existing   Delta   New
XGRAD    1.000      ----    1.000
YGRAD   -5.000      ----   -5.000
ZGRAD   -7.000      ----   -7.000
Accept the New Currents? (Y,N) [Y] : n
Exit LVshim? (Y,N) [N] : y
  
```

Figure 25

System Performance Testing (SPT)

1. From the Common Service Desktop PM page, click on "SPT PM Mode" in the expanded PM folder on the left side of the page. See Figure 26a and 26b.
2. To start the SPT tool, click on the link: "[Click here to start this tool](#)" shown in Figure 26a.

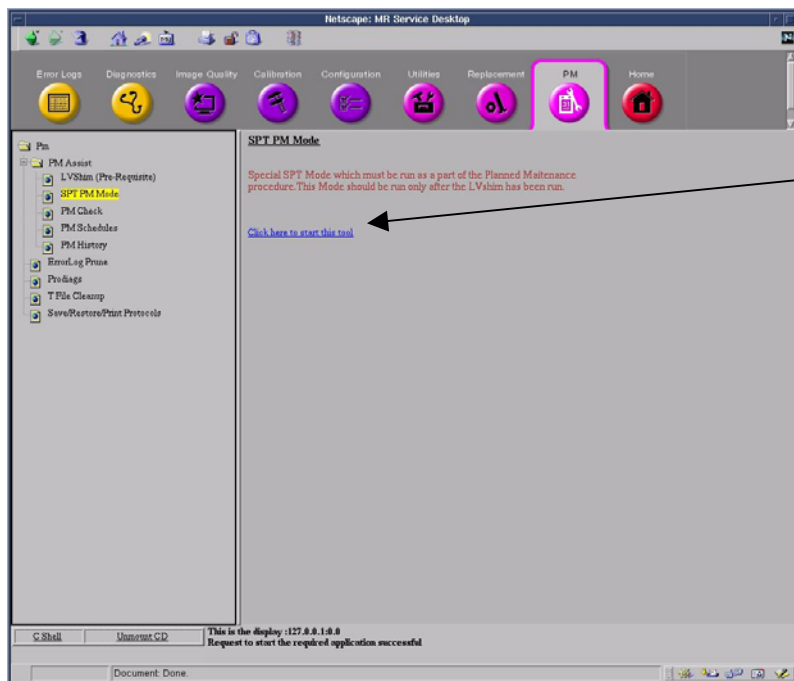


Figure 26a

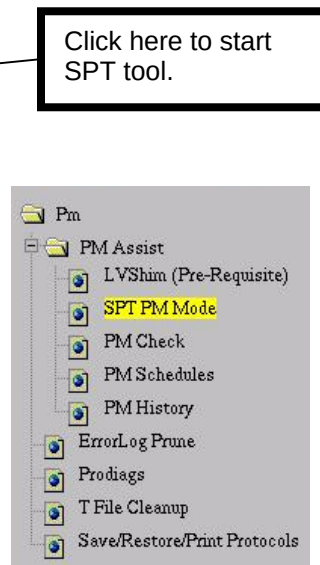


Figure 26b

3. When a window opens requesting selection of the Head Coil and Phantom type, click the appropriate radio button, and then click the [Start] button. See Figure 27.

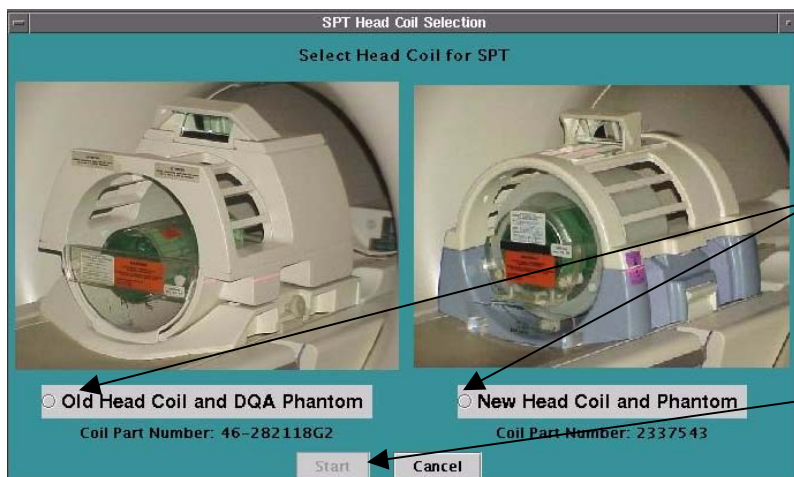


Figure 27

The System Performance Test (SPT) screen will open. See Figure 28a.

- For the PM SPT mode, all the tests are pre-selected and cannot be changed.
- The test takes just over one hour to complete. In the upper right corner of the SPT screen, the remaining test time will display. See Figures 28a and 28b.

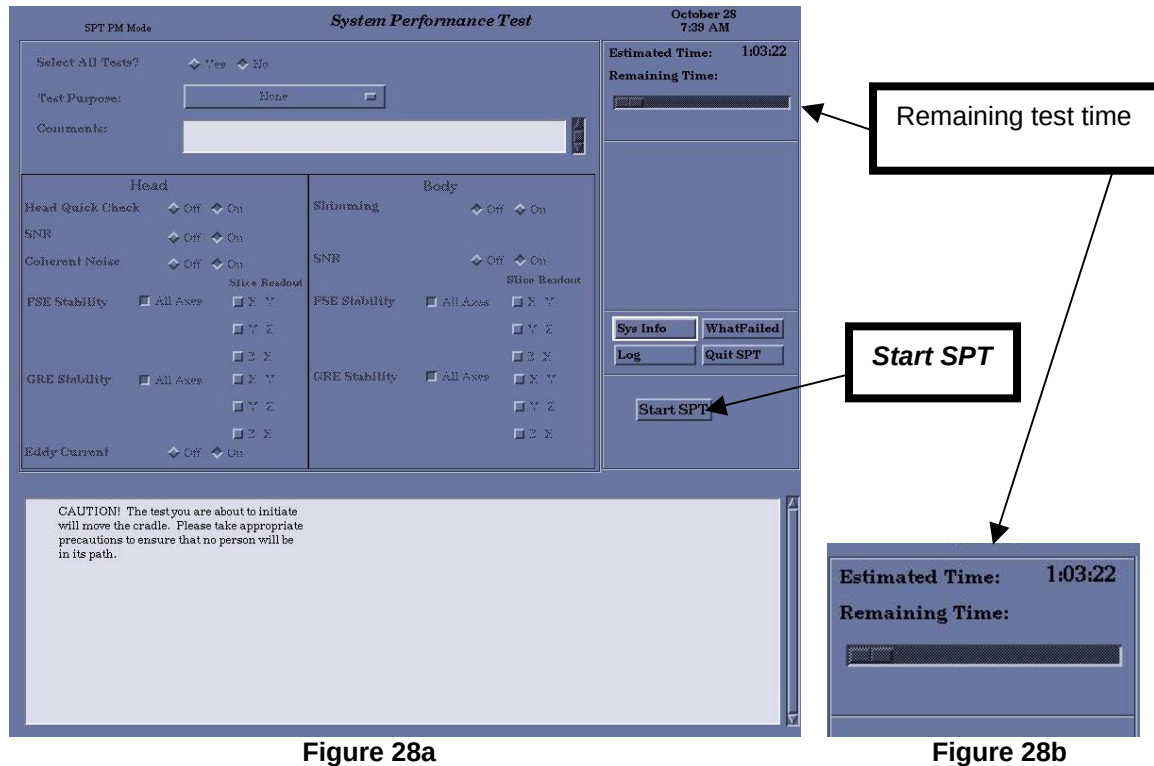


Figure 28a

Figure 28b

- Test Progress and Results will display in the lower window of the SPT page. See Figure 29.

System Performance Test

SPT PM Mode October 28 9:07 AM

Select All Tests? ☐ Yes ☐ No

Test Purpose:

Comments:

Head		Body	
Head Quick Check	☐ Off ☐ On	Shimming	☐ Off ☐ On
SNR	☐ Off ☐ On	SNR	☐ Off ☐ On
Coherent Noise	☐ Off ☐ On		
FSE Stability	☐ All Axes ☐ X ☐ Y ☐ Z	FSE Stability	☐ All Axes ☐ X ☐ Y ☐ Z
GRE Stability	☐ All Axes ☐ X ☐ Y ☐ Z	GRE Stability	☐ All Axes ☐ X ☐ Y ☐ Z
Eddy Current	☐ Off ☐ On		

Estimated Time:

Remaining Time:

Sys Info WhatFailed
Log Quit SPT
Abort SPT

Scanning eddy current (16 min 10 s)

CAUTION! The test you are about to initiate will move the cradle. Please take appropriate precautions to ensure that no person will be in its path.

Tue Oct 28 2003
08:56:00.660 Test Purpose: Planned Maintenance
08:56:01.241 Scanning first image for Head Quick Check (4 min 40 s)
08:57:05.812 Prescan is unsuccessful
08:57:06.011 Scanning eddy current (16 min 10 s)

Test Results

Figure 29

For more information on System Performance Tests, including phantom setup, consult *Direction 2333500, Signa Excite Service Methods, Troubleshooting / Image Quality Troubleshooting Tools / System Performance Tests (SPT)*.

Note: Failures MUST be resolved and the System Performance Test (SPT) MUST be re-run before proceeding to the PM Check.

PM Check

1. To verify that the PM was completed successfully, click on "PM Check" in the left column. See Figures 30a and 30b.

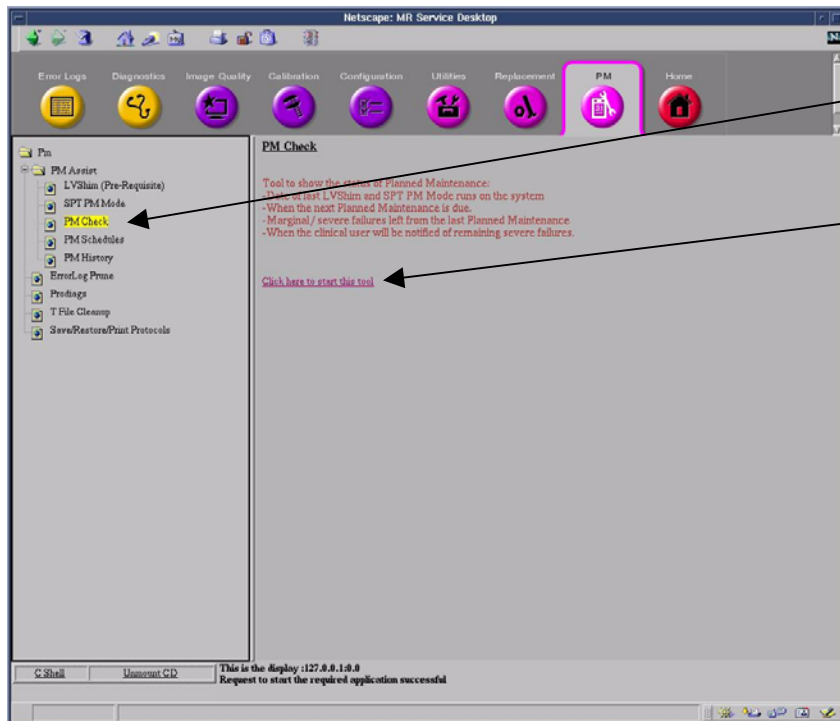


Figure 30a

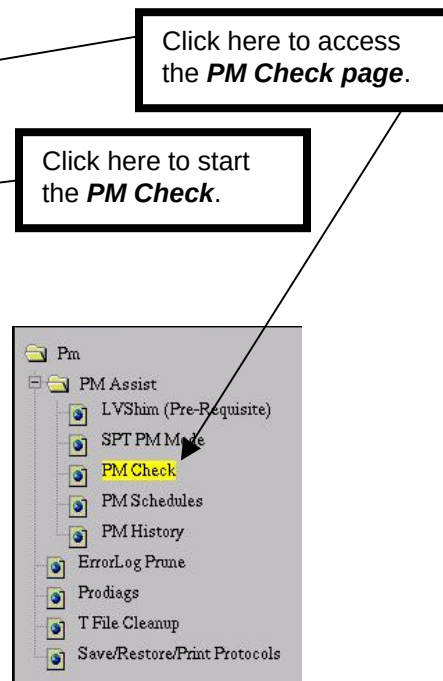


Figure 30b

2. Note the test results in the PM Check window that displays with the information shown in Figure 31.

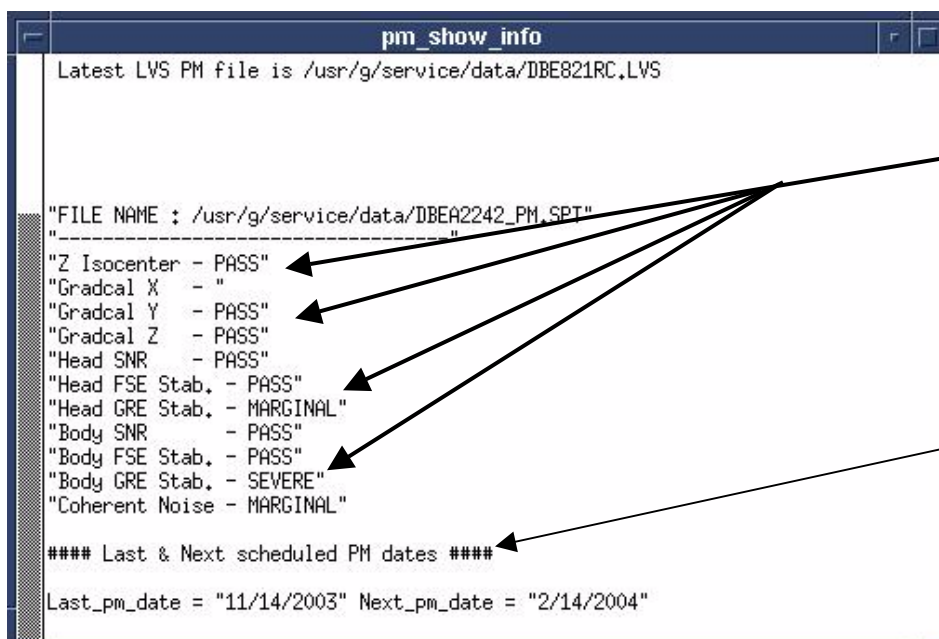


Figure 31

- As seen in the example shown in Figure 31, Body GRE Stab. has a SEVERE failure. This, and any other SEVERE issues, must be resolved in 21 days for the PM Assist to be considered complete. The system will notify the operator of the problem if not resolved within the 21-day period.
- For the PM Assist to be successful and complete, all test results must pass as shown in Figure 32.

```

pm_show_info
Latest LVS PM file is /usr/g/service/data/DB0848RC.LVS

"FILE NAME : /usr/g/service/data/DB0A1914_PM.SPT"
"-----"
"Z Isocenter - PASS"
"Gradcal X - PASS"
"Gradcal Y - PASS"
"Gradcal Z - PASS"
"Head SNR - PASS"
"Head FSE Stab. - PASS"
"Head GRE Stab. - PASS"
"Body SNR - PASS"
"Body FSE Stab. - PASS"
"Body GRE Stab. - PASS"
"Coherent Noise - PASS"

#### Last & Next scheduled PM dates ####
Last_pm_date = "11/24/2003" Next_pm_date = "2/24/2004"

```

Figure 32

PM Schedule

The PM Schedule Tool will show the following status:

- Date of Last LVShim and SPT PM Mode runs on the System
- When the next Planned Maintenance is due
- Marginal / Severe failures remaining from last Planned Maintenance
- When the clinical user will be notified of the remaining severe failures

1. Click on "PM Schedules" in the expanded PM folder. See Figures 33a and 33b. The right side of the screen will provide more details about PM Schedules and a link to start the PM Schedules tool.
2. Click on the link: "[Click here to start this tool](#)" (shown in Figure 33a) to start the PM Schedules tool.

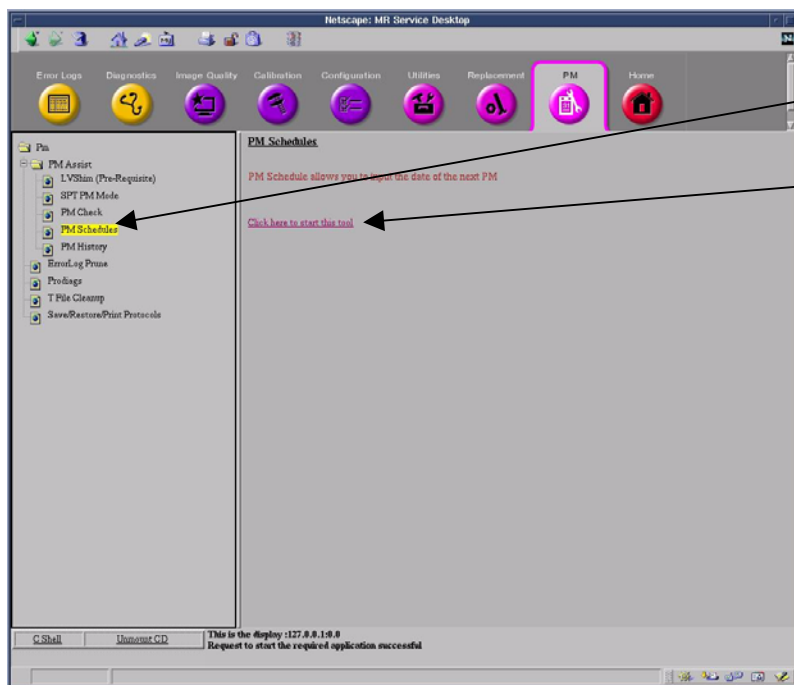


Figure 33a

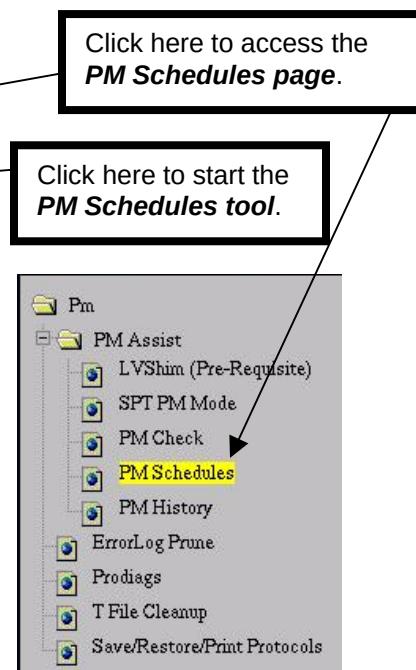


Figure 33b

PM Never Performed on This System

1. If Planned Maintenance has not been performed on this System, the following message displays in the tool:

There is no PM performed on this system. Schedule a date for PM, run LVShim & SPT PM test. Click on PM Check link on CSD to view results.

2. Enter your Name or initials, and press the [Enter] key. See Figure 34.

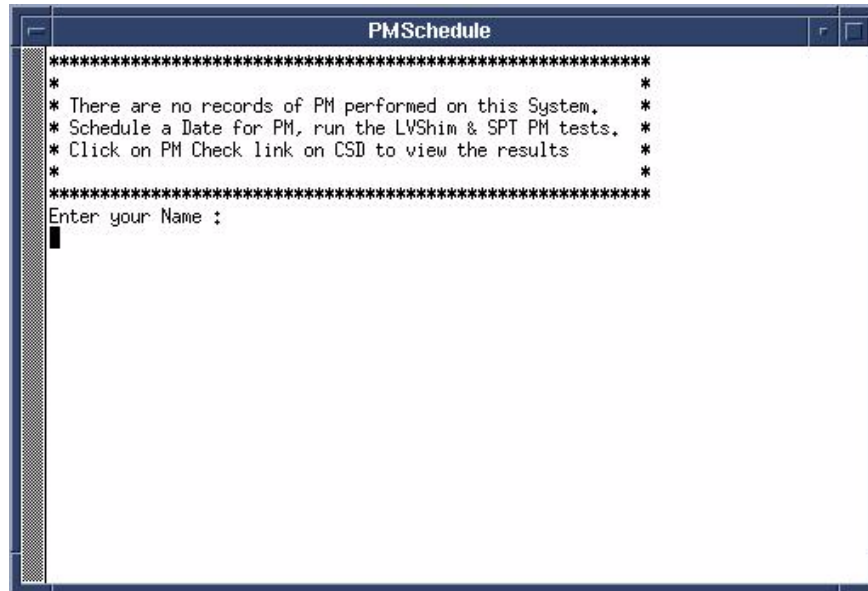


Figure 34

3. Enter the next scheduled PM date, and press the [Enter] key. See Figure 35.

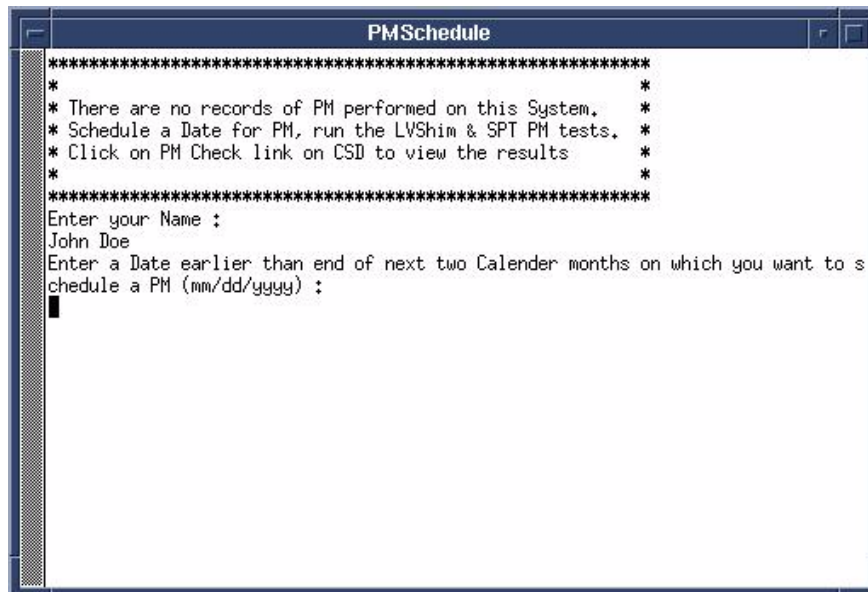
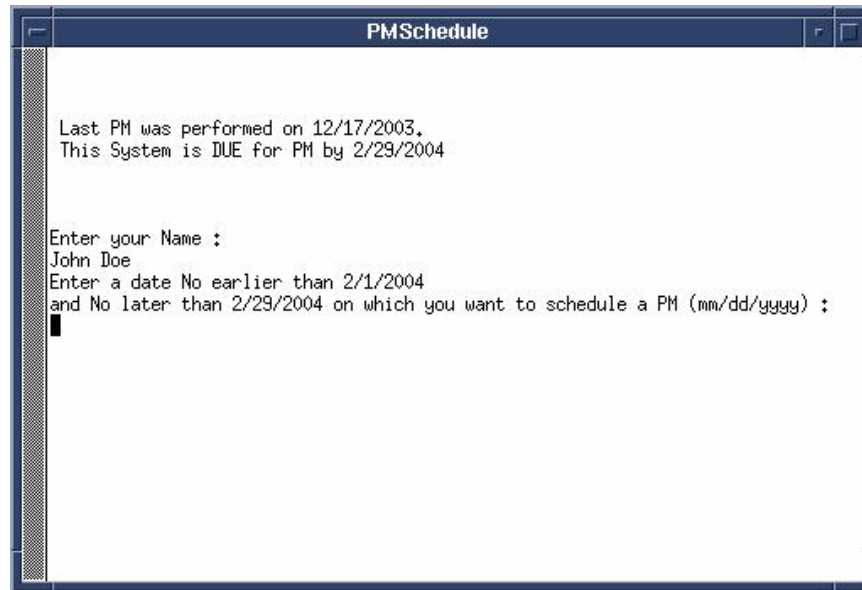


Figure 35

System Following Normal PM Schedule

The tool will display the when the last PM was performed and when the next PM is scheduled. See Figure 36.

1. Enter the date the next PM will be performed. Insert the date within the specified time limit provided by the tool. See Figure 36.



```

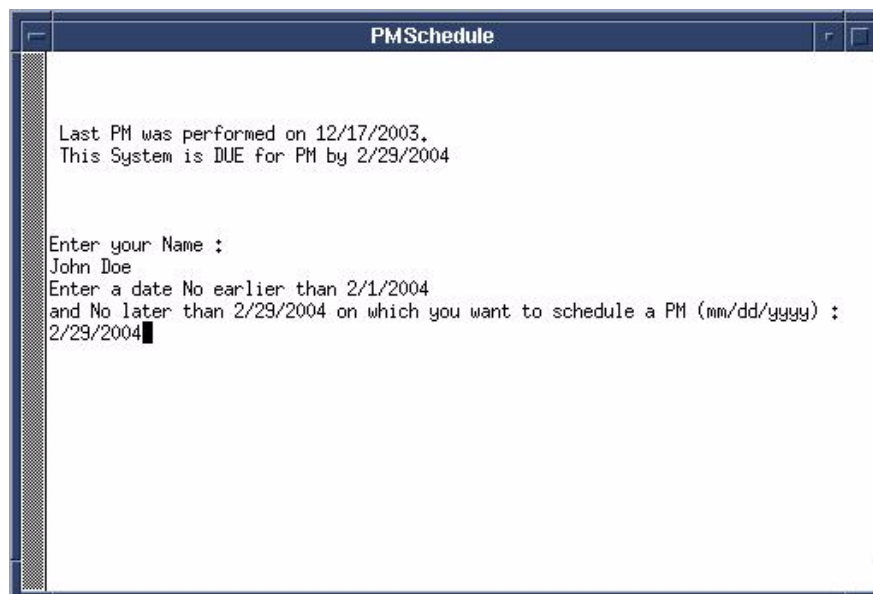
PMSchedule

Last PM was performed on 12/17/2003.
This System is DUE for PM by 2/29/2004

Enter your Name :
John Doe
Enter a date No earlier than 2/1/2004
and No later than 2/29/2004 on which you want to schedule a PM (mm/dd/yyyy) :
  
```

Figure 36

2. Press the [Enter] key. See Figure 37.



```

PMSchedule

Last PM was performed on 12/17/2003.
This System is DUE for PM by 2/29/2004

Enter your Name :
John Doe
Enter a date No earlier than 2/1/2004
and No later than 2/29/2004 on which you want to schedule a PM (mm/dd/yyyy) :
2/29/2004
  
```

Figure 37

PM History

1. Click on "PM History" in the expanded PM folder. See Figures 38a and 38b. The right side of the screen will provide more details about PM History and a link to start the PM History tool.
2. Click on the link: "[Click here to start this tool](#)" (shown in Figure 38a) to start the PM History tool.

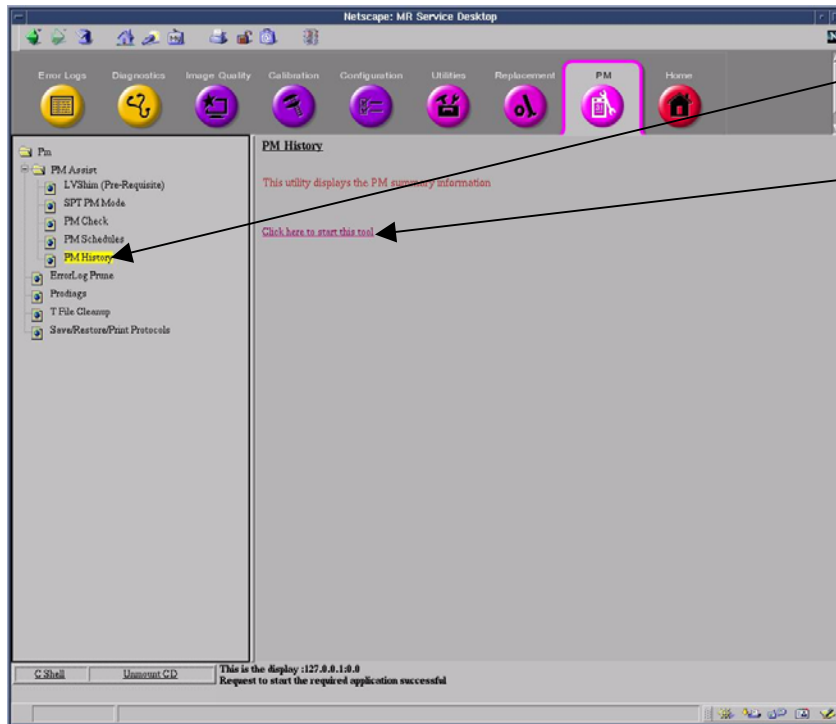


Figure 38a

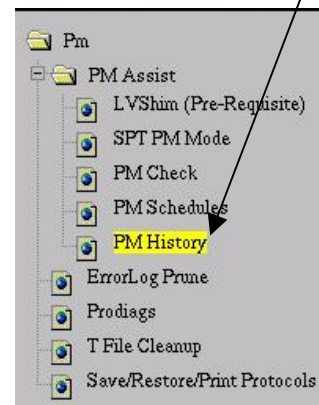


Figure 38b

3. The PM History (shown in Figure 39) displays the following information:

- Current Date
- Last Performed PM Date
- Month Next PM due.
- Customer Scheduled PM Date
- Field Engineer PM Date

Note: Customer Scheduled PM Date and Field Engineer PM Date should be the same date. If not, discuss PM scheduling with the customer immediately.

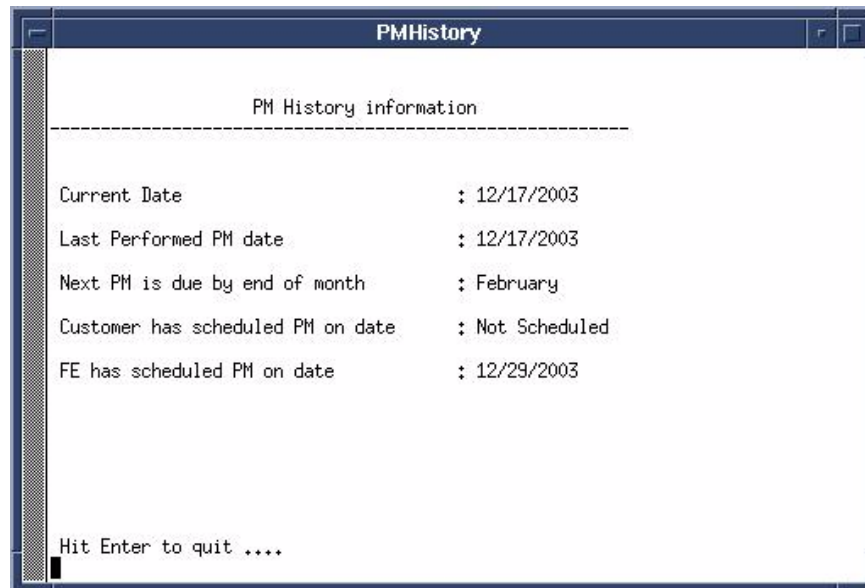


Figure 39

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